

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – TRANSFORMER MECHANICS WORKSHEET

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 3 – Transformer Mechanics Worksheet
Weightage:	40% of CIA	Total Marks:	100 (1 Question)
CO1 Statement:	<i>Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models. (BL1, BL2)</i>		
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Section / Batch:	BE-CSE-AI	Date:	28/07/2026

Instructions to Students

- Answer one questions. It carries 100 Marks. Total: 100 Marks.
- Tokenize a sentence.
- Convert tokens to token IDs.
- Explain embeddings.
- Label a Transformer encoder.
- Identify Query, Key, and Value.
- Explain which words receive higher attention.
- Compare Transformer models.
- Complete a Transformer pipeline.
- Draw a block diagram.

Code of Conduct

I sagar(192472070 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. Lakshmi sagar

SECTION A – TRANSFORMER MECHANICS WORKSHEET

12. Sentence Processing through Transformers Worksheet

PART A: CONCEPTUAL QUESTIONS — MODEL ANSWERS

Q1. Explain the concept of "self-attention" in a Transformer. Why is it more effective than the fixed context window used in traditional RNN/LSTM models? [3 Marks]

Answer:

Self-attention lets every token directly compare itself with every other token in the sentence in one step, producing a relevance-weighted blend of the whole sequence. RNNs/LSTMs instead pass context sequentially through a hidden state, so distant words must survive many steps before influencing each other — causing long-range information loss (vanishing gradient). Self-attention removes this distance penalty and is also fully parallelizable across the sequence.

Q2. Define the Query (Q), Key (K), and Value (V) vectors used in the self-attention mechanism. What role does each play? [3 Marks]

Answer:

Query (Q): what a word is "looking for" in the rest of the sentence.

Key (K): what each word "offers," compared against queries to compute relevance ($Q \cdot K$).

Value (V): the actual content that gets carried forward, weighted by the relevance score.

Q3. Why does the Transformer architecture require Positional Encoding? What problem would arise if it were removed? [2 Marks]

Answer:

Self-attention treats the input as an unordered set of tokens — it has no inherent sense of sequence. Without positional encoding, "dog bites man" and "man bites dog" would produce identical representations. Positional encoding (sinusoidal or learned) is added to each embedding so the model can distinguish word order.

Q4. Differentiate between "Multi-Head Attention" and "Single-Head Attention." State one advantage of using multiple heads. [3 Marks]

Answer:

Single-head attention produces one relevance/context pattern per token. Multi-head attention runs several attention operations in parallel, each with its own learned Q/K/V projections, then concatenates the results. Advantage: different heads can specialize in different relationships (e.g., syntax vs. coreference), giving a richer overall representation than a single head could capture.

Q5. Briefly describe the purpose of: (a) Add & Norm, (b) Feed-Forward Network. [2 Marks]

Answer:

(a) Add & Norm: adds a residual (skip) connection from the sub-layer's input to its output, then normalizes — stabilizes gradients and preserves the original signal through deep stacks.

(b) Feed-Forward Network: a position-wise fully connected network applied identically to every token, adding non-linearity and extra representational capacity after attention.

Q6. In the decoder, self-attention is "masked." Explain why masking is necessary during sentence generation. [2 Marks]

Answer:

The decoder must generate each word using only the words produced so far, not ones it hasn't generated yet. Masking sets attention scores for future positions to $-\infty$ before softmax, forcing zero attention to them and preserving the autoregressive (left-to-right) generation order.

PART B: NUMERICAL / CALCULATION — WORKED SOLUTION

Q7. Compute the raw attention scores for token w_1 (I) against w_1 , w_2 and w_3 . [4 Marks]

Answer:

$$\text{score}(Q_1, K_1) = (1 \times 1 + 0 \times 1) / \sqrt{2} = 1 / 1.41 = 0.71$$

$$\text{score}(Q_1, K_2) = (1 \times 1 + 0 \times 0) / \sqrt{2} = 1 / 1.41 = 0.71$$

$$\text{score}(Q_1, K_3) = (1 \times 0 + 0 \times 1) / \sqrt{2} = 0 / 1.41 = 0.00$$

Q8. Apply Softmax to the three scores from Q7. [3 Marks]

Answer:

$$\exp(0.71) = 2.03, \exp(0.71) = 2.03, \exp(0) = 1.00 \rightarrow \text{sum} = 5.07$$

$$\alpha_{11} = 2.03 / 5.07 = 0.40$$

$$\alpha_{12} = 2.03 / 5.07 = 0.40$$

$$\alpha_{13} = 1.00 / 5.07 = 0.20$$

Q9. Compute the final context-aware output vector Z_1 for token w_1 . [3 Marks]

Answer:

$$\begin{aligned} Z_1 &= 0.40 \times [2, 0] + 0.40 \times [0, 2] + 0.20 \times [1, 1] \\ &= [0.8, 0] + [0, 0.8] + [0.2, 0.2] \end{aligned}$$

$$Z_1 = [1.0, 1.0]$$

Q10. Interpret what Z_1 represents about the word "I" in this sentence. [1 Marks]

Answer:

Z_1 shows that "I" is now represented as a roughly equal blend of itself and "love" (its two highest attention weights, 0.40 each), with a smaller contribution from "AI" — the model has updated "I"'s representation mainly using context from the verb next to it.

PART C: DIAGRAM-BASED — GUIDANCE FOR EVALUATION

Q11. Complete Transformer architecture — expected diagram elements. [5 Marks]

Answer:

Encoder stack (left): Input Embedding $\rightarrow$ + Positional Encoding $\rightarrow$ [Multi-Head Self-Attention $\rightarrow$ Add & Norm $\rightarrow$ Feed-Forward $\rightarrow$ Add & Norm] $\times N$

Decoder stack (right): Output Embedding (shifted right) $\rightarrow$ + Positional Encoding $\rightarrow$ [Masked Multi-Head Self-Attention $\rightarrow$ Add & Norm $\rightarrow$ Multi-Head Cross-Attention (over encoder output) $\rightarrow$ Add & Norm $\rightarrow$ Feed-Forward $\rightarrow$ Add & Norm] $\times N \rightarrow$ Linear $\rightarrow$ Softmax

Award marks for: correct block order, correct residual/norm placement, and correct labelling of the encoder-decoder cross-attention link.

Q12. Self-attention matrix for "AI processes sentences" — expected shading. [3 Marks]

Answer:

Highest attention weight for "processes" is typically expected on itself and on "AI" (its subject) and/or "sentences" (its object) — i.e., the cells (processes, AI) and/or (processes, sentences) in the 3×3 grid, since a verb's representation is most influenced by the words it grammatically connects to.

Accept any well-justified shading, as exact weights depend on the (untrained) model in this exercise — grade on the quality of the justification, not one "correct" cell.

PART D: SENTENCE-PROCESSING WALKTHROUGH — MODEL ANSWERS

Stage	Expected answer
1. Tokenization	Sentence is split into tokens/sub-words: [Transformers, understand, context, .]
2. Input Embedding	Each token is mapped to a dense numeric vector representing its meaning.
3. Positional Encoding added	A position-specific vector is added to each embedding so word order (1st, 2nd, 3rd...) is encoded.
4. Self-Attention (Q,K,V)	Each token's embedding is projected into Q, K, V vectors; attention scores are computed between all token pairs.
5. Multi-Head Attention output	Several attention heads run in parallel and their outputs are concatenated into one context-aware vector per token.
6. Add & Norm + Feed-Forward	Residual connection + normalization stabilize the signal; a feed-forward network adds further non-linear transformation per token.
7. Decoder / Output generation	The decoder uses masked self-attention plus cross-attention to the encoder output to generate the next word/token, one step at a time, via Linear + Softmax.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical / Concept	30%	All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Application	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Quality	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Presentation	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Documentation / Communication	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technical / Concept(30%)	Application(25%)	Quality (20%)	Presentation (15%)	Documentation / Communication (10%)	Total Marks (100)
Q1						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____